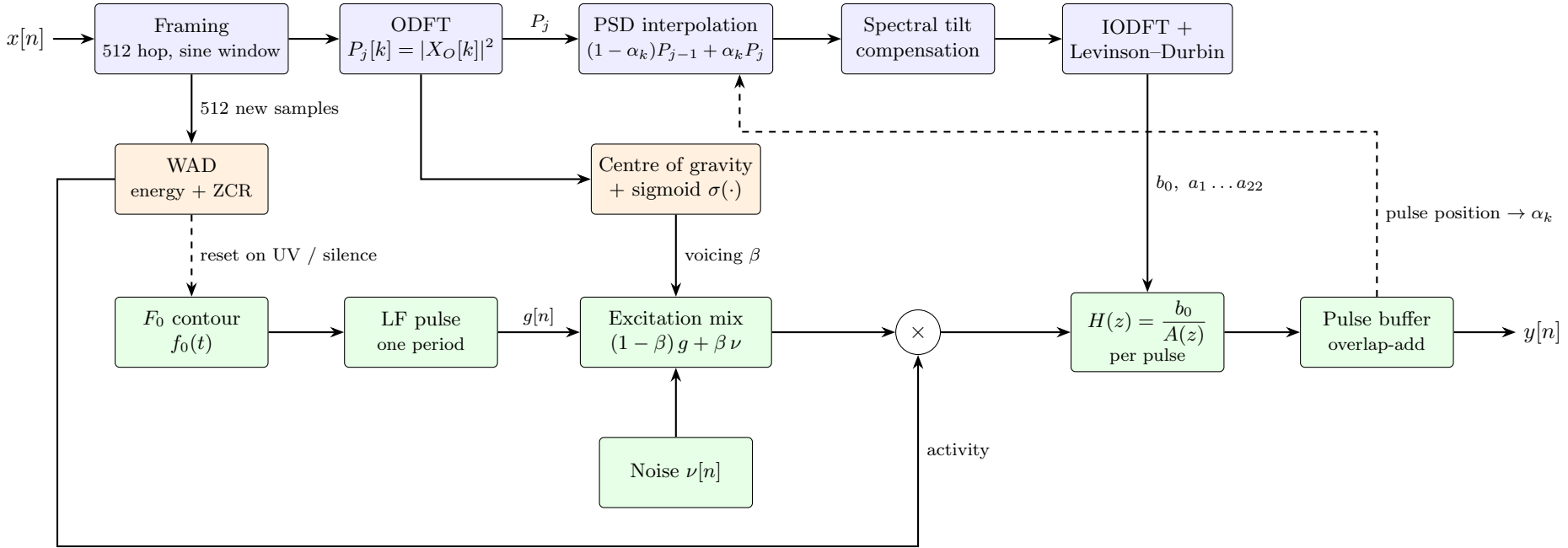

$x[n]$
Framing
512 hop, sine window
ODFT
$P_j[k] = |X_O[k]|^2$
P_j
PSD interpolation
$(1-\alpha_k)P_{j-1} + \alpha_k P_j$
Spectral tilt
compensation
IODFT +
Levinson–Durbin
512 new samples
WAD
energy + ZCR
Centre of gravity
+ sigmoid $\sigma(\cdot)$
$b_0,\ a_1 \ldots a_{22}$
pulse position $\rightarrow \alpha_k$
reset on UV / silence
voicing β
F_0 contour
$f_0(t)$
LF pulse
one period
$g[n]$
Excitation mix
$(1-\beta)\, g + \beta\, \nu$
$\times$
$H(z) = \dfrac{b_0}{A(z)}$
per pulse
Pulse buffer
overlap-add
$y[n]$
Noise $\nu[n]$
activity